Assignment — FastAPI + PostgreSQL + React

1. Create PostgreSQL database

```
CREATE DATABASE product_store;
```

Connect to `product_store`.

2. Create `products` table

```
CREATE TABLE products (
    id INT PRIMARY KEY,
    name VARCHAR(100),
    price DECIMAL(10,2),
    image VARCHAR(500)
);
```

3. Insert product data into PostgreSQL

```
INSERT INTO products VALUES
(
    1,
    'Wireless Headphones',
    4500,
    'https://images.unsplash.com/photo-1505740420928-5e560c06d30e'
),
(
    2,
    'Smart Watch',
    7000,
    'https://images.unsplash.com/photo-1523275335684-37898b6baf30'
),
(
    3,
    'Laptop',
    65000,
```

```
    'https://images.unsplash.com/photo-1496181133206-80ce9b88a853'
);
```

---

4. Install PostgreSQL driver in FastAPI project

```
python -m pip install psycopg2-binary
```

---

5. Modify FastAPI

Remove the hardcoded:

```
products = [...]
```

Create a PostgreSQL connection:

```
import psycopg2


def get_connection():
    return psycopg2.connect(
        host="localhost",
        database="product_store",
        user="postgres",
        password="YOUR_PASSWORD"
    )
```

---

6. GET products from PostgreSQL

Create:

```
@app.get("/products")
def get_products():

    conn = get_connection()
    cursor = conn.cursor()
```

```
    cursor.execute("""
        SELECT id, name, price, image
        FROM products
        ORDER BY id
    """)

    rows = cursor.fetchall()

    cursor.close()
    conn.close()

    products = []

    for row in rows:
        products.append({
            "id": row[0],
            "name": row[1],
            "price": float(row[2]),
            "image": row[3]
        })

    return products
```

The API should return:

```
[
  {
    "id": 1,
    "name": "Wireless Headphones",
    "price": 4500,
    "image": "https://..."
  }
]
```

---

7. POST should insert into PostgreSQL

Ask them to modify:

```
POST /products
```

so it performs:

```
INSERT INTO products
```

instead of:

```
products.append(...)
```

---

8. PUT should update PostgreSQL

Ask them to implement:

```
PUT /products/{product_id}
```

using:

```
UPDATE products
SET name = ...,
    price = ...,
    image = ...
WHERE id = ...
```

---

9. DELETE should delete from PostgreSQL

Implement:

```
DELETE /products/{product_id}
```

using:

```
DELETE FROM products
WHERE id = ...
```

---

10. React stays the same

React should still call:

```
fetch("http://127.0.0.1:8000/products")
```

React should not care whether FastAPI gets the data from:

```
Python List
```

or:

```
PostgreSQL
```